

Compensating Wage Differentials

ECON 300 - Labour Economics

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Learning Objectives

- Explain how wages reflect job attributes: working conditions, shift work, pensions, health benefits, and risk.
- Identify when market-determined workplace risk can be optimal and when it will not be.
- Evaluate how safety regulations can raise or lower overall welfare.
- Understand “no free lunch”: workers often **pay** for desirable perks via lower wages.
- Use compensating wage premia for risk to infer the *value of a statistical life* (VSL).

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- What determines relative pay rates across jobs?
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Purposes of Wages and Wage Structures

Wage Structure

A hierarchy/grid of wage levels; a set of **premiums** linked to worker or job attributes.

- **Wage–productivity link:** Pay must cover marginal productivity (in competitive settings).
- **Social role:** Wages also signal status, fairness, and help align incentives.

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Smith's Compensating Differentials: What Gets Priced?

- Agreeableness/disagreeableness of tasks
- Training cost and difficulty
- Turnover/retention aspects
- Responsibility, power, trust
- Probability of success
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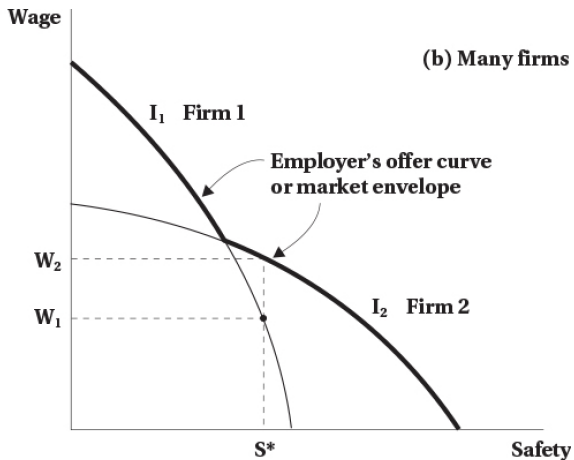
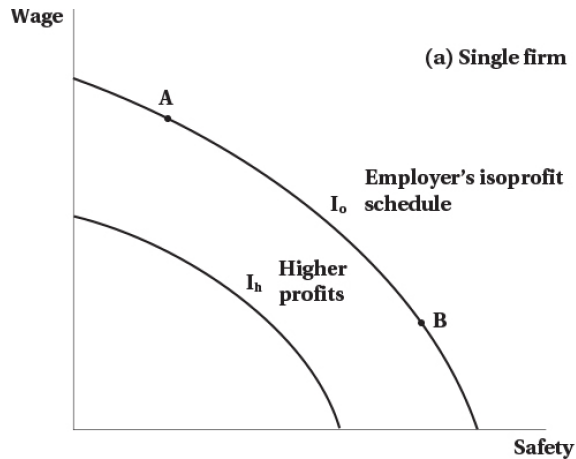
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- Combinations of **wage** and **safety** that yield the same profit.
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Figure 8.1: Employer's Isoprofit Schedules and Offer Curve



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- Firms differ in the **cost** of providing safety.
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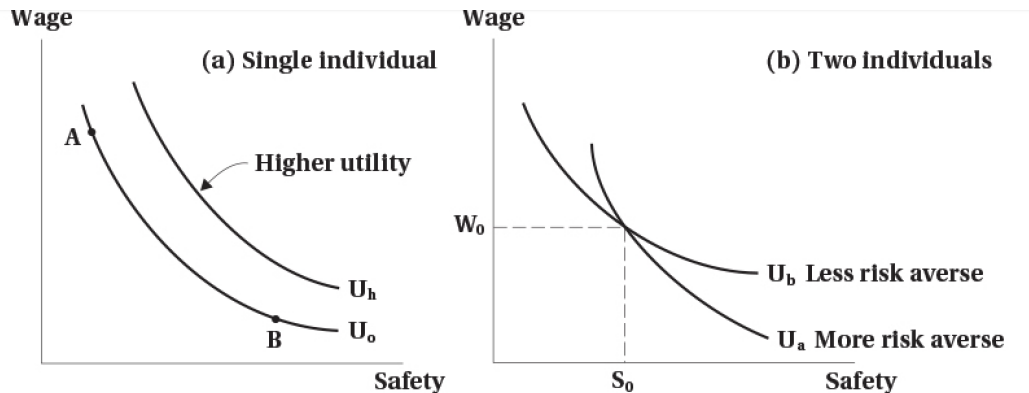
Worker Preferences and Indifference Curves

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Figure 8.2: Worker Indifference Curves



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- **Tangency** of iso-utility and isoprofit $\Rightarrow$ optimal wage–safety bundle.
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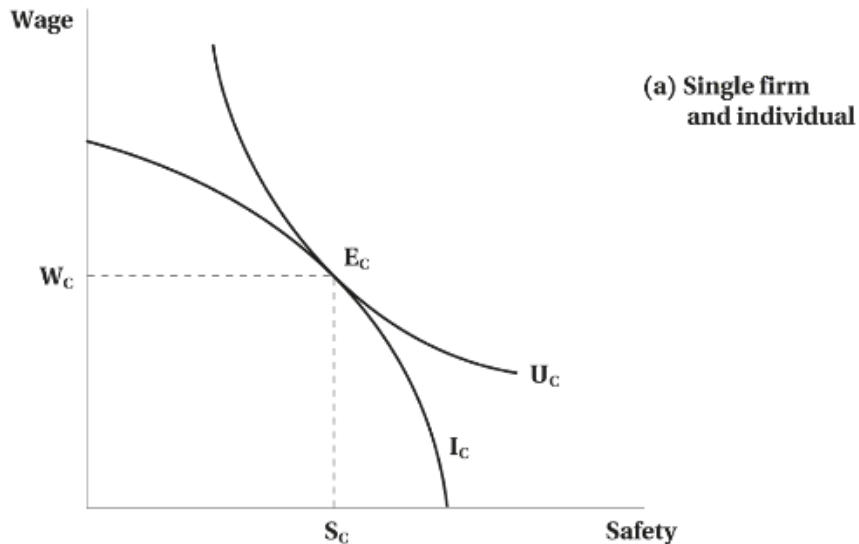
Many Firms, Many Workers (Sorting)

- With competition, workers **sort** into jobs by risk preferences; firms sort by safety technology.
- **Wage–safety locus:** Market envelope of equilibrium wage–safety combinations.

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Figure 8.3: Market Equilibrium



Characteristics of the Wage–Safety Locus

- **Negative slope:** Lower safety requires compensating wages.
- **Curvature** depends on worker preferences and firm safety technology.

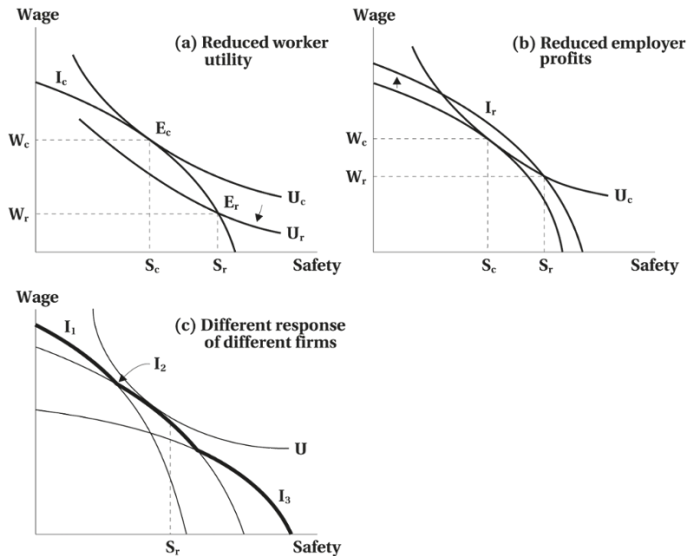
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Figure 8.4: Responses to a Safety Standard



Worked Example: Attracting Doctors to Rural Areas

Two doctors can earn \$100,000 in a city.

- Dr. Brown's city amenity value is \$90,000 $\Rightarrow$ indifferent between \$100,000 (city) and \$190,000 (rural).
- Dr. Lewis's amenity value is \$40,000 $\Rightarrow$ indifferent between \$100,000 (city) and \$140,000 (rural).
- **Implication:** Rural premia must differ across workers to induce relocation; sorting yields higher rural pay.

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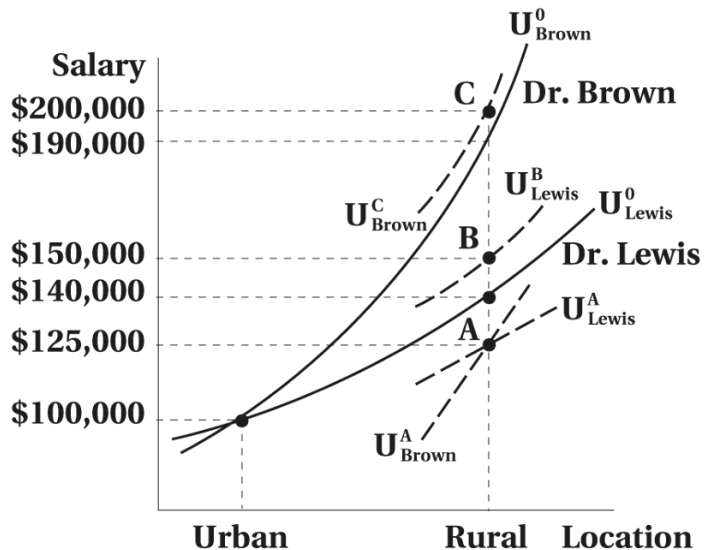
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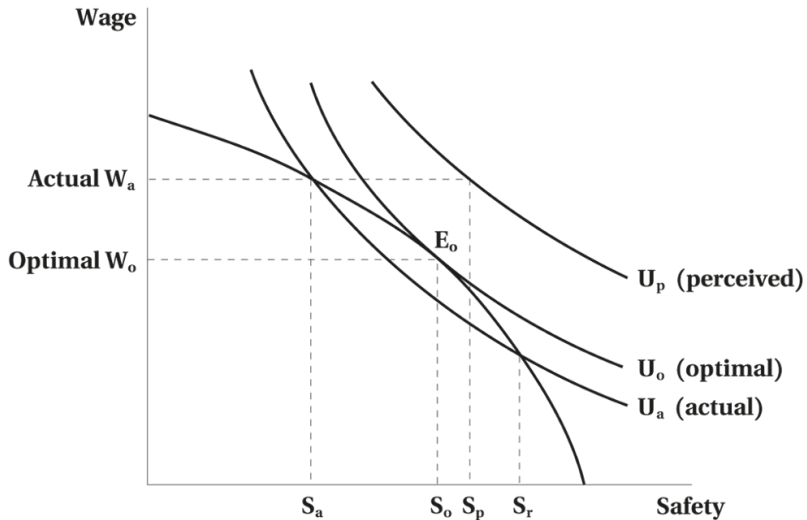
Location Decision Illustration



- If workers misperceive risk, mandated safety can raise their utility *and* keep employers whole.
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Figure 8.5: Effect of Imperfect Information



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- Information failures and non-competitive markets
- External costs of accidents not borne by workers/firms alone
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- Errors-in-variables (risk mismeasurement)
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- Sample selection (who takes risky jobs)

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